D3-Induced Error Analysis for Selected Oxide Materials

Question 1

Table 1: Materials Project IDs and space groups for the selected materials.

Material	Materials Project ID	Space Group
NiO	mp-19009	Fm-3m
CoO	mp-22408	F-43m
FeO	mp-1274279	C2/m
TiO_2	mp-390	$I4_1/amd$
RuO_2	mp-825	$P4_2/mnm$
IrO_2	mp-2723	$P4_2/mnm$
$SrTiO_3$	mp-4651	I4/mcm
$LaMnO_3$	mp-17554	Pnma

Question 2

In this project, the D3-induced error is called ΔE. It measures how much the D3 dispersion energy curve drops below a smooth expected baseline during cell expansion.

Protocol

1. Start from the relaxed Materials Project structure.
2. Generate a set of uniformly expanded or strained cells:

$$a_i = a_0(1 + s_i)$$

   Here, a_0 is the original cell length scale, s_i is the applied strain, and a_i is the strained cell length.

3. For each strained structure, run the D3 calculation and record:

$$\text{cell_length} = x_i$$

$$\text{D3 dispersion energy} = E_{\text{dis},i}$$

4. Plot E_{dis} against cell length.

5. Locate the abnormal dip in the curve. The dip point is:

$$x_{\text{dip}} = \text{cell length at the detected dip}$$
$$E_{\text{min}} = E_{\text{dis}}(x_{\text{dip}})$$

6. We estimate the smooth expected D3 value using a local peak-dip-recovery method. First, we choose the local pre-dip shoulder (x_L, E_L), then we choose the recovered post-dip point (x_R, E_R). We draw a straight local baseline between these two points:

$$E_{\text{expected}}(x) = mx + b$$
$$m = \frac{E_R - E_L}{x_R - x_L}$$
$$b = E_L - mx_L$$

Here, (x_L, E_L) is the local left shoulder before the artificial drop, and (x_R, E_R) is the first recovered point after the dip. This local baseline represents the smooth D3 behavior expected if the coordination switching artifact had not occurred.

7. Evaluate the local baseline at the dip position:

$$E_{\text{expected,dip}} = E_{\text{expected}}(x_{\text{dip}})$$

$E_{\text{expected,dip}}$ is therefore the predicted smooth dispersion energy at the same cell length as the observed dip. It is not another D3 calculation; it is the baseline value obtained by interpolation between the local shoulder and local recovery points.

8. Define the D3-induced error:

$$\Delta E = E_{\text{min}} - E_{\text{expected,dip}}$$

Variable Definitions

Table 2: Variables in the local D3-induced error calculation.

Variable	Meaning
x_i	Cell length of the ith strained structure.
s_i	Strain applied to generate the ith structure.
$E_{\text{dis},i}$	D3 dispersion energy calculated by `s-dftd3` for the ith strained structure.
x_L	Cell length at the local left shoulder before the artificial dip.
E_L	D3 dispersion energy at the local left shoulder.
x_R	Cell length at the recovered post-dip point.
E_R	D3 dispersion energy at the recovered post-dip point.
m	Slope of the local baseline between (x_L, E_L) and (x_R, E_R).
b	Intercept of the local baseline.
x_{dip}	Cell length where the artificial D3 dip is detected.
E_{min}	Observed D3 dispersion energy at x_{dip}; this is the bottom of the dip.
$E_{\text{expected,dip}}$	Smooth baseline-predicted D3 energy at x_{dip}; this is what the energy should be without the artificial dip.
ΔE	D3-induced error, calculated as $E_{\text{min}} - E_{\text{expected,dip}}$.
$\lvert\Delta E\rvert$	Magnitude of the D3-induced error.

9. Interpret the result:

- If $\Delta E < 0$, the D3 curve has an artificial negative dip.
- A more negative ΔE means a larger D3-induced error.
- If ΔE is close to zero, there is no strong D3-induced dip by this protocol.

Schematic

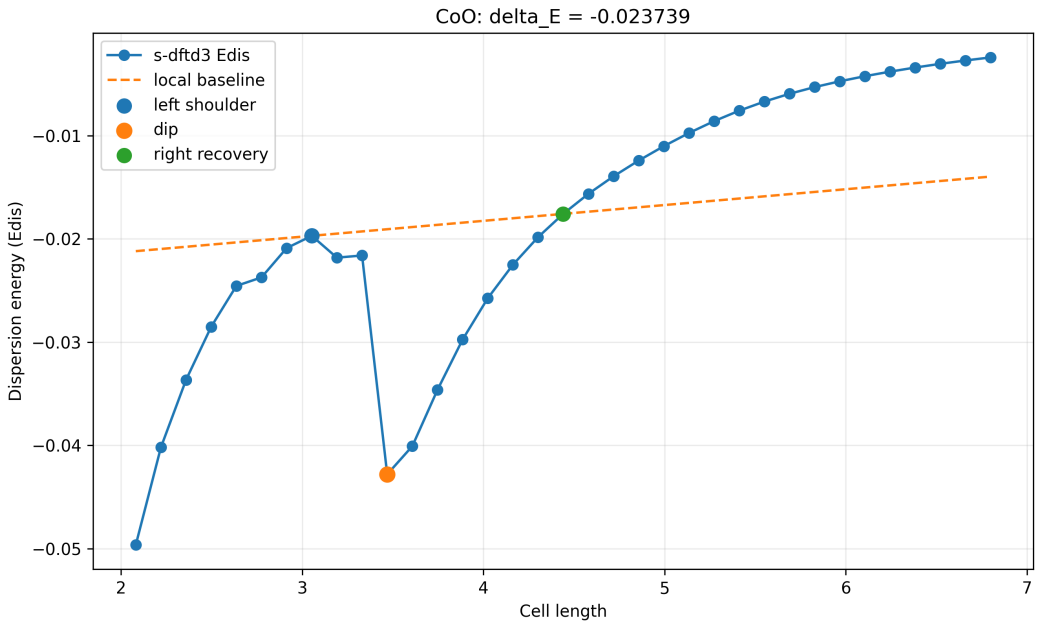


Figure 1: Example local-baseline construction for CoO. The D3-induced error is calculated as $\Delta E = E_{\text{min}} - E_{\text{expected,dip}}$, where the dip point lies below the expected local baseline.

Final Reported Values

For each material, we report x_{dip}, E_{min}, $E_{\text{expected,dip}}$, and ΔE. The magnitude $|\Delta E|$ is the measured D3-induced error.

Question 3

We calculated the D3-induced error from the strained `s-dftd3` dispersion-energy curves using the local peak-dip-recovery baseline method. For each material, we drew the baseline from the local pre-dip shoulder to the recovered post-dip point, and calculated the error as:

$$\Delta E = E_{\text{min}} - E_{\text{expected,dip}}$$

Negative ΔE means the D3 curve drops below the expected smooth baseline. The absolute value $|\Delta E|$ is the magnitude of the D3-induced error.

Table 3: D3-induced error values for all eight materials.

Material	Materials Project ID	Space Group	Dip Cell Length	E_{min}	$E_{\text{expected,dip}}$	ΔE	$\lvert\Delta E\rvert$
NiO	mp-19009	Fm-3m	3.204498	-0.031086	-0.012859	-0.018227	0.018227
CoO	mp-22408	F-43m	3.468908	-0.042808	-0.019069	-0.023739	0.023739
FeO	mp-1274279	C2/m	2.720810	-0.091038	-0.046713	-0.044324	0.044324
TiO_2	mp-390	$I4_1$/amd	5.144189	-0.033192	-0.013372	-0.019820	0.019820
RuO_2	mp-825	$P4_2$/mnm	3.422476	-0.047043	-0.038392	-0.008651	0.008651
IrO_2	mp-2723	$P4_2$/mnm	4.288553	-0.033895	-0.030374	-0.003522	0.003522
$SrTiO_3$	mp-4651	I4/mcm	7.174065	-0.100512	-0.038759	-0.061753	0.061753
$LaMnO_3$	mp-17554	Pnma	8.513574	-0.222176	-0.089542	-0.132634	0.132634

Error Ranking

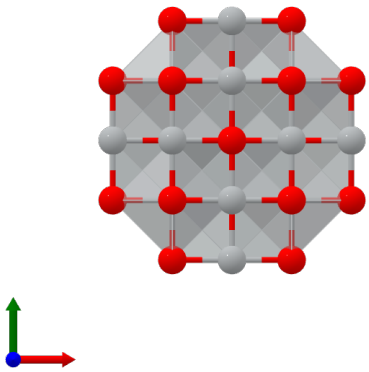
Table 4: Ranking from largest to smallest D3-induced error magnitude.

Rank	Material	$\lvert\Delta E\rvert$
1	$LaMnO_3$	0.132634
2	$SrTiO_3$	0.061753
3	FeO	0.044324
4	CoO	0.023739
5	TiO_2	0.019820
6	NiO	0.018227
7	RuO_2	0.008651
8	IrO_2	0.003522

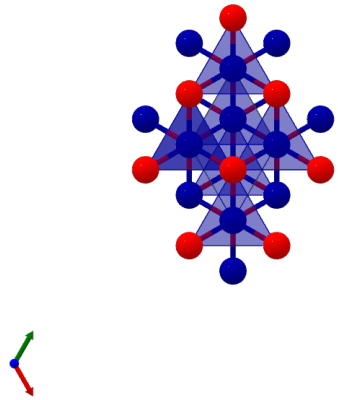
Therefore, $LaMnO_3$ shows the largest D3-induced error among the eight tested materials, while IrO_2 shows the smallest.

Structure Gallery

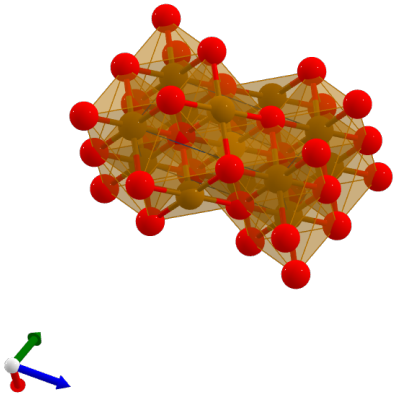
Before analyzing the D3-error trend, we show the crystal structures of all eight materials. These structures help connect the numerical ΔE values to the local geometry, packing, and coordination environments discussed in Question 4.



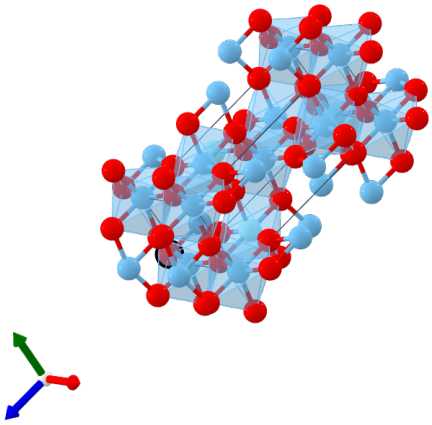
NiO: compact rocksalt structure, Fm-3m.



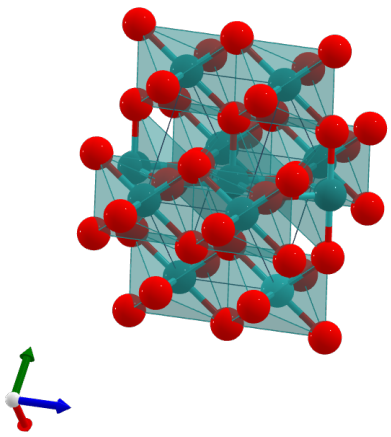
CoO: rocksalt-like monoxide structure, R3m view.



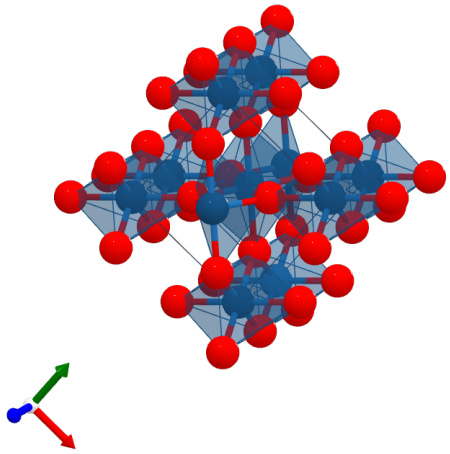
FeO: distorted monoxide structure with lower symmetry.



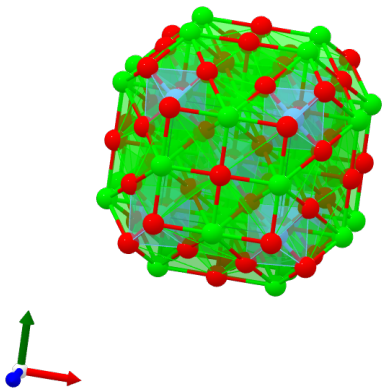
TiO_2: anatase-type TiO_6 octahedral network.



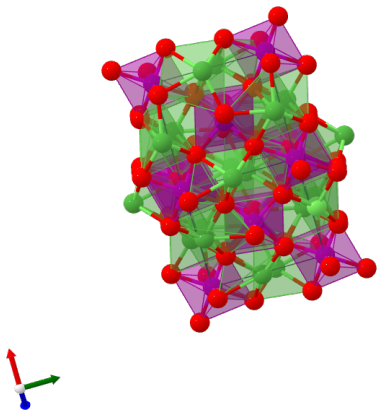
RuO_2: compact rutile-type octahedral framework.



IrO_2: compact rutile-type framework with smooth coordination.



$SrTiO_3$: perovskite framework with large Sr A-site cations.



$LaMnO_3$: distorted perovskite with non-equivalent Mn–O environments.

Question 4

The calculated D3-induced error magnitudes follow the order:

$$LaMnO_3 > SrTiO_3 > FeO > CoO > TiO_2 > NiO > RuO_2 > IrO_2$$

This trend shows that the D3-induced artifact is not controlled by a single factor such as atomic mass, oxidation state, or elemental polarizability. Instead, the error mainly depends on how the local coordination environment changes during expansion. Since D3 calculates dispersion coefficients using coordination-number-dependent interpolation, a sharp change in coordination number can produce an abrupt change in C_6, leading to an artificial dip in the dispersion-energy curve.

Therefore, the D3 error is best understood as a **geometry-driven coordination artifact**.

Structure-Type Dependence

The largest errors occur in the perovskite-type oxides:

$$LaMnO_3 > SrTiO_3$$

This is reasonable because perovskites contain multiple chemically distinct sublattices: a large A-site cation, a smaller B-site transition-metal cation, and an extended oxygen octahedral framework. During cell expansion, several A–O, B–O, and O–O distances change at different rates. As a result, different atomic pairs may cross the D3 coordination-number switching region at different strain values.

This makes the D3 C_6 interpolation highly non-smooth and produces a large artificial stabilization.

$$\text{small change in distance} \rightarrow \text{sharp CN change} \rightarrow \text{abrupt } C_6 \rightarrow \text{D3 artifact}$$

Among the two perovskites, $LaMnO_3$ gives the largest error. This can be attributed to its distorted Pnma structure, large La-site cation, and Jahn–Teller-active Mn^{3+}. These features create multiple non-equivalent La–O and Mn–O environments, making the coordination-number response more irregular during expansion.

$SrTiO_3$ also shows a large error because the large Sr site interacts with many oxygen neighbors at relatively long distances. Such weak, long-range contacts are especially sensitive to D3 coordination cutoffs.

Monoxide Trend

The monoxide systems lie in the intermediate error range:

$$FeO > CoO > NiO$$

This trend shows that simple binary composition alone does not guarantee a small D3 artifact.

NiO has a relatively small error because it is compact and highly symmetric. Its rocksalt-like structure provides a regular local coordination environment, so the coordination number changes more smoothly during expansion.

FeO, however, shows a much larger error than NiO. This is because the selected FeO structure is monoclinic rather than an ideal cubic rocksalt structure. The lower symmetry creates non-equivalent Fe–O environments. During strain, these environments do not change uniformly, so the D3 coordination number can change more abruptly.

In addition, Fe^{2+} is generally more polarizable than Ni^{2+}, and this can amplify the magnitude of the dispersion correction. Thus, FeO combines both structural distortion and electronic softness, making it more vulnerable to D3 artifacts.

Rutile Oxides

The smallest errors are observed for rutile-type oxides:

$$RuO_2 > IrO_2$$

At first glance, this may seem surprising because Ru and Ir are heavy elements, and heavier atoms are often more polarizable. One might expect larger dispersion-related errors. However, the results show the opposite: IrO_2 has the smallest D3-induced error.

This proves an important point:

$$\text{D3 artifact} \neq \text{heavy element effect only}$$

The reason RuO_2 and IrO_2 have small errors is their compact and regular rutile structure. The metal–oxygen octahedral network changes smoothly during expansion. Since the local coordination environment does not undergo abrupt switching, the D3 C_6 coefficients also vary smoothly.

Thus, even though Ru and Ir are heavy and polarizable, the regular rutile geometry suppresses the D3 artifact.

Ionic Size and Packing

The larger errors in $LaMnO_3$ and $SrTiO_3$ are also consistent with their relatively open perovskite frameworks. A larger volume per atom generally means that atoms have more space to move during expansion, and weak longer-range contacts may enter or leave the coordination environment.

This is especially important for large A-site cations such as La^{3+} and Sr^{2+}. These ions interact with many oxygen atoms over relatively long distances. Such contacts are not strongly covalent, so their D3 coordination contributions can switch sharply as the structure expands.

However, packing alone is not sufficient to explain the full trend. For example, CoO and TiO_2 have similar volume-per-atom values, but their D3 errors are different. This means the local bonding topology, symmetry, and coordination switching behavior are more important than volume alone.

Oxidation State Effects

Oxidation state does not show a simple direct correlation with D3 error.

For example, TiO_2, RuO_2, and IrO_2 all contain formally tetravalent transition metals, but their errors are much smaller than $LaMnO_3$ and $SrTiO_3$. This indicates that oxidation state is secondary to geometry and local coordination behavior.

TiO_2 has a relatively small error because Ti^{4+} is a hard d^0 cation with lower polarizability. Its Ti–O network is also more regular than the distorted perovskite systems. Therefore, the D3 dispersion correction changes more smoothly.

In contrast, $LaMnO_3$ contains Mn^{3+}, which is Jahn–Teller active. This creates distorted MnO_6 octahedra and multiple unequal bond lengths. Such structural distortion enhances coordination-number sensitivity and leads to the largest D3-induced error.

Covalent or Regular Frameworks

D3 artifacts are reduced when the bonding environment is stable and regular. In covalent or compact frameworks, nearest neighbors are well defined and bond distances change smoothly. As a result, the coordination number does not abruptly switch between reference environments.

This suppresses artificial jumps in C_6 and leads to smoother dispersion-energy curves.

The key point is:

error depends on instability of CN, not simply number of neighbors

A material can have many neighbors and still show a small D3 error if the coordination environment changes smoothly. Conversely, a material with fewer atoms can show a large error if distortion causes abrupt coordination switching.

Main Anomalies

IrO_2

IrO_2 is an important anomaly because Ir is heavy and polarizable, yet IrO_2 shows the smallest error. This demonstrates that elemental polarizability alone does not control the artifact. The compact rutile structure keeps the coordination environment smooth, suppressing the D3 error.

FeO

FeO has a larger error than TiO_2, NiO, RuO_2, and IrO_2 despite being a simple binary oxide. This is because the selected FeO structure is monoclinic and less symmetric than ideal rocksalt. The distorted Fe–O coordination environment makes the D3 correction more sensitive to strain.

$LaMnO_3$

$LaMnO_3$ is the strongest outlier. Its large error arises from the combined effect of an open perovskite lattice, a large La cation, Jahn–Teller-active Mn^{3+}, and multiple non-equivalent bond distances. These factors make coordination switching especially strong.

Overall Conclusion

The observed trend can be summarized as:

$$\text{distorted perovskites} > \text{distorted monoxides} > \text{compact rutile oxides}$$

The D3-induced error is largest in structures with open packing, large ions, multiple sublattices, distorted coordination polyhedra, and strain-sensitive bonding environments. It is smallest in dense, regular, and smoothly varying structures.

Therefore, the D3 error is not simply a function of formula, metal mass, oxidation state, or polarizability. It is primarily controlled by how sharply the local coordination environment changes during structural expansion.

Final takeaway:

$$\boxed{\text{D3 artifact is mainly a coordination-switching problem.}}$$

All Local Delta Plots

In this final section, we present the local-baseline delta plots for all eight materials. Each plot shows the calculated D3 dispersion-energy curve, the local baseline, the left shoulder, the detected dip, and the recovered post-dip point used to calculate ΔE.

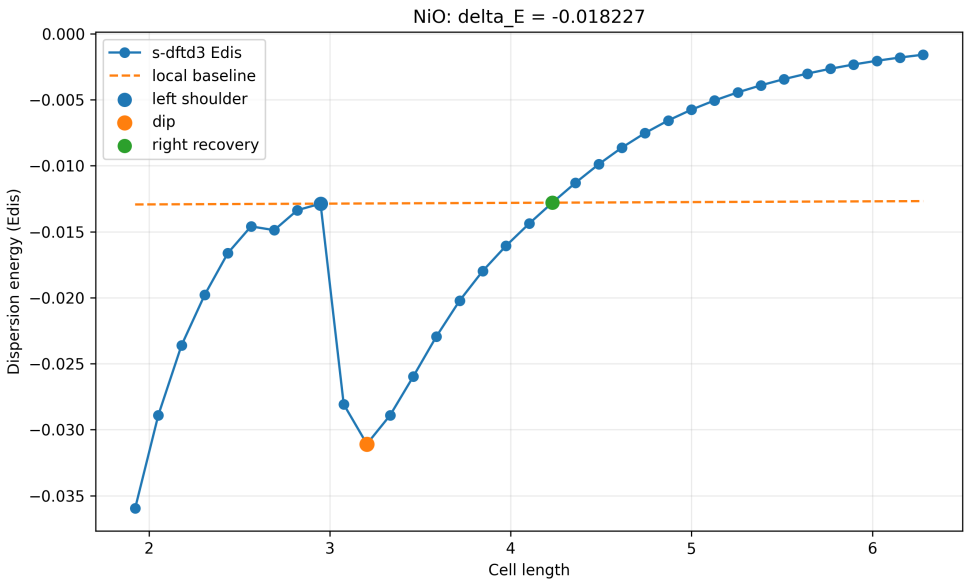


Figure 2: Local D3-induced error plot for NiO.

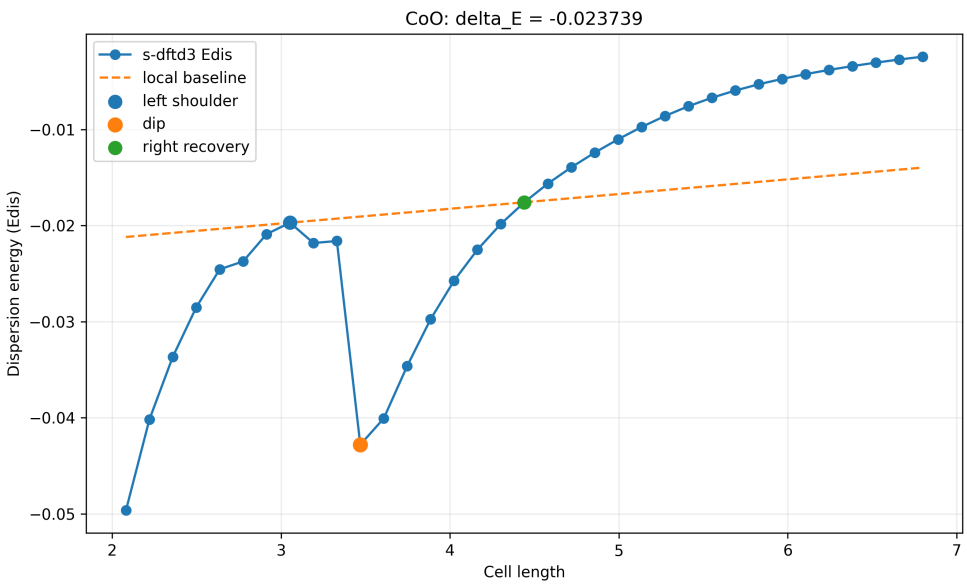


Figure 3: Local D3-induced error plot for CoO.

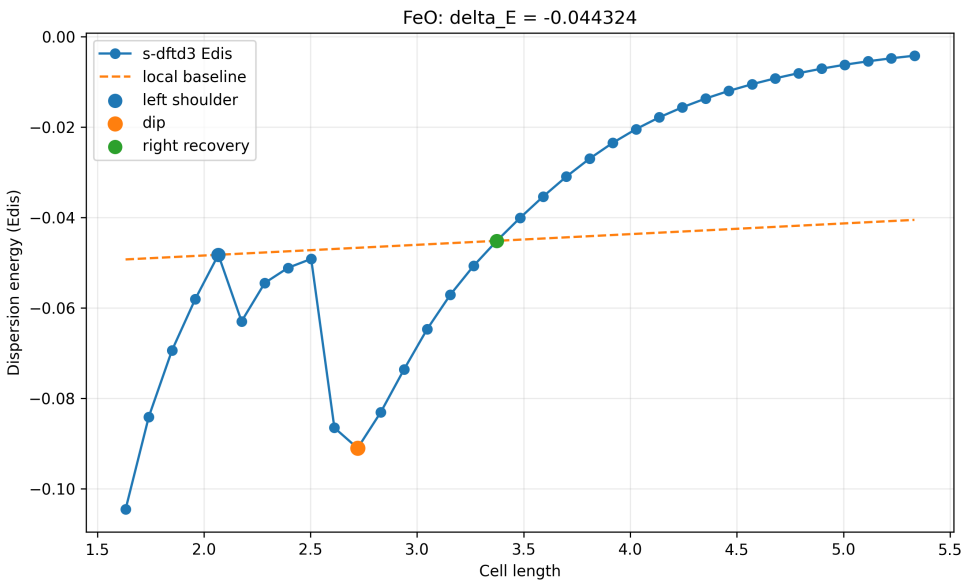


Figure 4: Local D3-induced error plot for FeO.

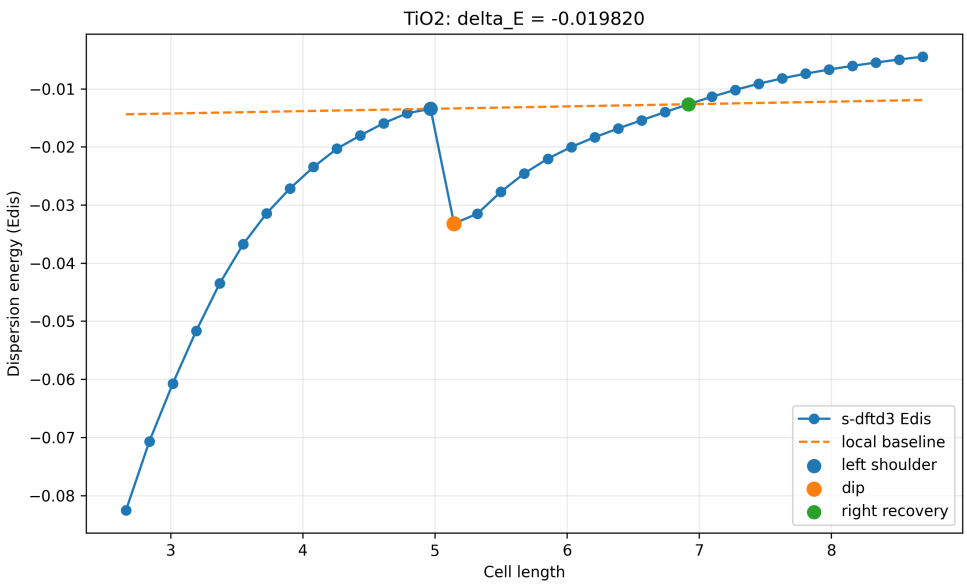


Figure 5: Local D3-induced error plot for TiO_2.

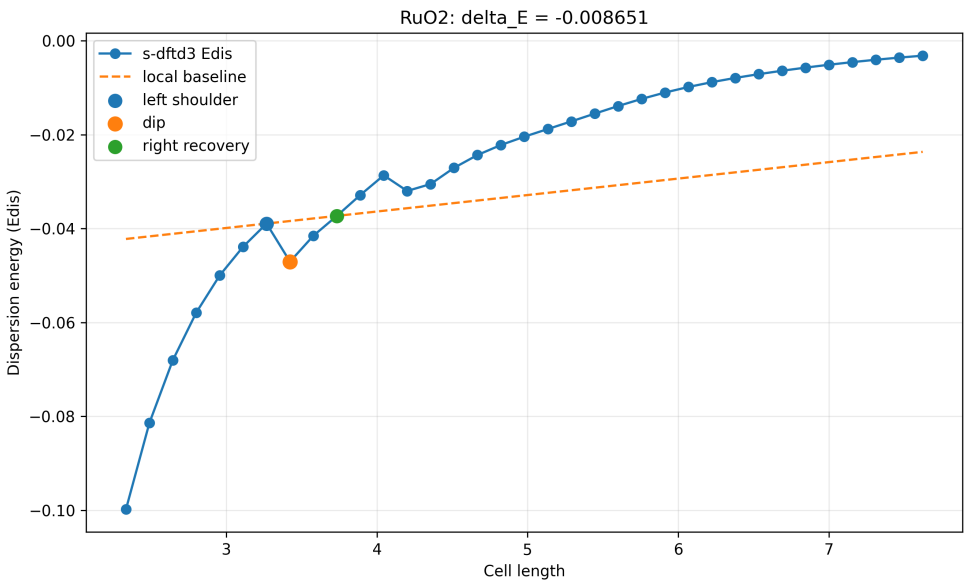


Figure 6: Local D3-induced error plot for RuO_2.

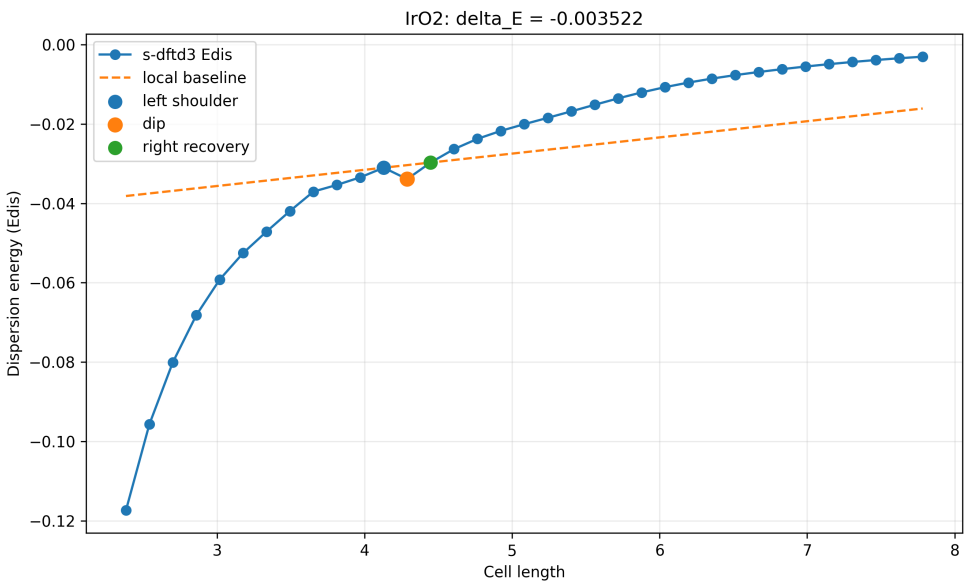


Figure 7: Local D3-induced error plot for IrO_2.

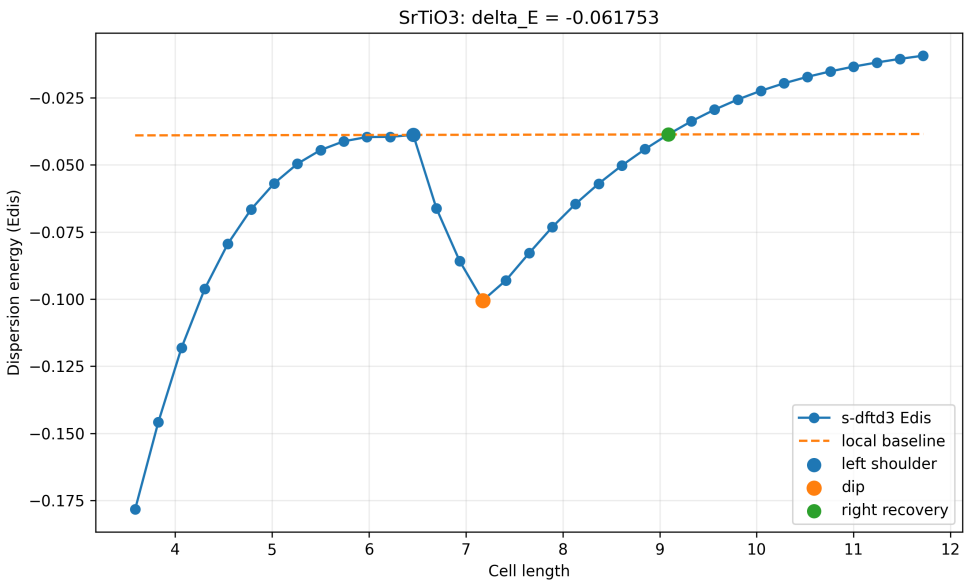


Figure 8: Local D3-induced error plot for $SrTiO_3$.

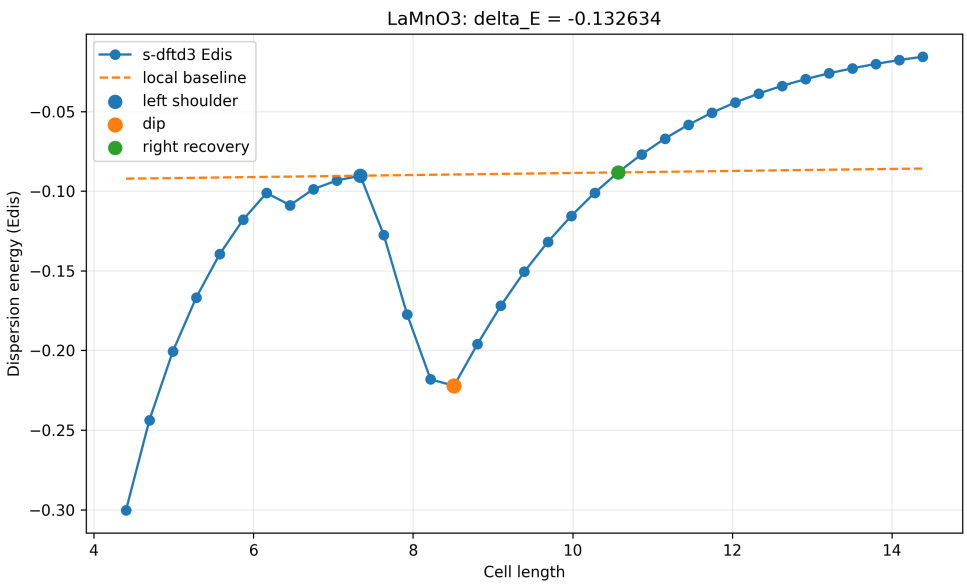


Figure 9: Local D3-induced error plot for $LaMnO_3$.